Respiration

Original exam-style questions · 20 questions

Original questions written with AI tools. Report anything that looks wrong. Topical Hub is independent and is not affiliated with or endorsed by Cambridge University Press & Assessment.

Question 1

core · 1 mark · 0610_T12_P1_20261013_Q1

Which list contains only processes that use energy released by respiration?

A diffusion, growth, muscle contraction

B active transport, cell division, protein synthesis

C growth, osmosis, passage of nerve impulses

D active transport, diffusion, maintaining a constant body temperature

Question 2

core · 1 mark · 0610_T12_P1_20261013_Q2

Which statement describes aerobic respiration?

A chemical reactions in cells that make energy from oxygen

B the exchange of oxygen and carbon dioxide across a gas exchange surface

C the movement of air into and out of the lungs

D chemical reactions in cells that use oxygen to break down nutrient molecules to release energy

Topical Hub
topical-hub.pages.dev

Question 3

core · 1 mark · 0610_T12_P1_20261013_Q3

Some germinating bean seeds are placed in a sealed flask. The flask is kept in the dark at 25 °C. The seeds respire aerobically.

After 24 hours, the air in the flask is analysed.

Which row shows how the concentrations of oxygen and carbon dioxide in the air have changed?

	oxygen	**carbon dioxide**
A	decreased	increased
B	increased	decreased
C	decreased	decreased
D	increased	increased

Question 4

supplement · 1 mark · 0610_T12_P1_20261013_Q4

Which is the balanced chemical equation for aerobic respiration?

A $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

B $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 12H_2O$

C $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

D $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

Question 5

core · 1 mark · 0610_T12_P1_20261013_Q5

Which statement describes anaerobic respiration?

A chemical reactions in cells that break down nutrient molecules to release energy without using oxygen

B chemical reactions in cells that use oxygen to break down nutrient molecules but release less energy

C the breakdown of nutrient molecules that happens only in organisms living where there is no oxygen

D the process in the lungs that releases energy when breathing stops

Question 6

core · 1 mark · 0610_T12_P1_20261013_Q6

Which row describes anaerobic respiration in human muscles, compared with aerobic respiration?

	oxygen used	**energy released per glucose molecule**	**product or products**
A	no	more	lactic acid
B	no	much less	carbon dioxide and water
C	yes	much less	lactic acid
D	no	much less	lactic acid

Topical Hub
topical-hub.pages.dev

Question 7

core · 1 mark · 0610_T12_P1_20261013_Q7

Yeast cells are placed in a glucose solution in a sealed container with no oxygen.

Which substances are produced by the respiration of the yeast?

A lactic acid only

B alcohol and carbon dioxide

C carbon dioxide and water

D lactic acid and carbon dioxide

Question 8

core · 1 mark · 0610_T12_P1_20261013_Q8

During the sprint finish of a race, the leg muscles of a cyclist cannot get enough oxygen. Some of the muscle cells start to respire anaerobically.

Which word equation shows this anaerobic respiration?

A glucose → alcohol + carbon dioxide

B glucose + oxygen → carbon dioxide + water

C glucose → lactic acid

D glucose → lactic acid + carbon dioxide

Question 9

supplement · 1 mark · 0610_T12_P1_20261013_Q9

Some yeast cells respire anaerobically. They produce 40 molecules of ethanol, C_2H_5OH.

How many molecules of glucose were broken down, and how many molecules of carbon dioxide were produced?

	molecules of glucose broken down	**molecules of carbon dioxide produced**
A	20	40
B	40	40
C	20	120
D	80	80

Topical Hub
topical-hub.pages.dev

Question 10

supplement · 1 mark · 0610_T12_P1_20261013_Q10

After a race, a runner continues to breathe deeply and quickly for several minutes. The runner has an oxygen debt.

Which statement explains the cause of the oxygen debt?

A Carbon dioxide has built up in the lungs and must be breathed out.

B Lactic acid has built up in the muscles and blood, and oxygen is needed for its aerobic respiration.

C All the glucose in the muscles has been used up and must be replaced.

D Alcohol has built up in the muscles and must be removed by the kidneys.

Question 11

core · 6 marks · 0610_T12_P1_20261013_Q11

A student investigates the effect of temperature on respiration in yeast.

She mixes 10 cm^3 of yeast suspension with 10 cm^3 of glucose solution in a test tube. The test tube is closed with a bung. A delivery tube leads from the bung into a beaker of water. She places the test tube in a water bath and leaves it for 5 minutes to reach the temperature of the water bath. She then counts the number of bubbles that come out of the delivery tube in 5 minutes.

She repeats the experiment at different temperatures, using fresh yeast and glucose solution each time.

The table shows her results.

temperature / °C	**number of bubbles in 5 minutes**
10	8
20	21
30	46
40	72
50	25
60	0

(a) Name the gas in the bubbles. [1]

(b) Describe the effect of temperature on the number of bubbles produced. Use data from the table in your answer. [3]

(c) Calculate the mean number of bubbles produced per minute at 30 °C. [1]

(d) Suggest why no bubbles were produced at 60 °C. [1]

Topical Hub
topical-hub.pages.dev

Question 12

core · 5 marks · 0610_T12_P1_20261013_Q12

Living organisms use the energy released by respiration for many processes.

(a) State two uses of energy in the human body, other than muscle contraction. [2]

(b) Root hair cells take in mineral ions from the soil, even when the concentration of the ions in the soil is lower than the concentration inside the cells.

State the name of this process. [1]

(c) Mice keep their body temperature constant. Scientists found that a mouse kept at 5 °C respired faster than the same mouse kept at 25 °C.

Suggest why the mouse respired faster at 5 °C. [2]

Question 13

core · 5 marks · 0610_T12_P1_20261013_Q13

Two vacuum flasks, P and Q, are set up in the dark at 20 °C.

- Flask P contains 50 germinating pea seeds.
- Flask Q contains 50 pea seeds that have been boiled and then cooled.

Air is pumped slowly through each flask and then bubbled through a test tube of limewater. Before it enters each flask, the air passes through a chemical that removes all the carbon dioxide from it. Each flask also contains a thermometer.

After 30 minutes, the limewater connected to flask P has turned milky. The limewater connected to flask Q is still clear.

(a) State the word equation for aerobic respiration. [2]

Topical Hub
topical-hub.pages.dev

(b) Explain why the limewater connected to flask P turned milky but the limewater connected to flask Q stayed clear. [2]

(c) After two days, the temperature in flask P has risen but the temperature in flask Q has not changed.

Suggest why the temperature in flask P rose. [1]

Question 14

supplement · 5 marks · 0610_T12_P1_20261013_Q14

Liver cells respire aerobically.

(a) Write the balanced chemical equation for aerobic respiration. [2]

(b) In one experiment, a sample of liver cells uses 1.2 × 10^8 molecules of oxygen in aerobic respiration.

Calculate:

(i) the number of glucose molecules broken down [1]

(ii) the number of water molecules produced. [1]

Topical Hub
topical-hub.pages.dev

(c) State the ratio of the number of molecules of carbon dioxide produced to the number of molecules of oxygen used in aerobic respiration. [1]

Question 15

core · 5 marks · 0610_T12_P1_20261013_Q15

Rice plants are often grown in flooded fields. The soil around the roots is waterlogged and contains very little oxygen. For part of the time, the root cells respire anaerobically.

(a) Describe anaerobic respiration. [2]

(b) State how the energy released per glucose molecule by anaerobic respiration compares with the energy released by aerobic respiration. [1]

(c) Root cells take in mineral ions by active transport, using energy from respiration.

Suggest why the root cells take in mineral ions more slowly when the soil is waterlogged. [2]

Question 16

core · 6 marks · 0610_T12_P1_20261013_Q16

A baker mixes flour, water, sugar and yeast to make bread dough. The dough is left in a warm place. The yeast respires anaerobically. The baker measures the volume of the dough every 15 minutes.

time / min	**volume of dough / cm³**
0	200
15	236
30	278
45	314
60	340

(a) State the word equation for anaerobic respiration in yeast. [2]

(b) Explain why the volume of the dough increased. [2]

Topical Hub
topical-hub.pages.dev

(c) Calculate the percentage increase in the volume of the dough between 0 minutes and 60 minutes. [2]

Question 17

core · 5 marks · 0610_T12_P1_20261013_Q17

During a 100 m sprint, some of the muscle cells in the legs of an athlete respire anaerobically.

(a) State the word equation for anaerobic respiration in muscles. [1]

(b) Suggest why some muscle cells respire anaerobically during a 100 m sprint. [2]

(c) During a long-distance race, the muscle cells of the athlete respire mainly aerobically. The breathing rate and heart rate of the athlete are higher than when the athlete is resting.

Suggest how the higher breathing rate and heart rate help the muscle cells to keep respiring aerobically. [2]

Topical Hub
topical-hub.pages.dev

Question 18

supplement · 7 marks · 0610_T12_P1_20261013_Q18

An athlete runs as fast as possible on a treadmill from 0 minutes to 10 minutes. The athlete then sits and rests from 10 minutes to 50 minutes. The concentration of lactic acid in the blood of the athlete is measured.

time / min	lactic acid concentration in blood / mmol per dm^3
0	1.0
5	6.5
10	11.0
20	7.4
30	4.4
40	2.0
50	1.1

(a) Explain why the lactic acid concentration in the blood increased between 0 minutes and 10 minutes. [2]

(b) At 10 minutes the athlete has an oxygen debt.

Outline how the oxygen debt is removed after the exercise. [3]

(c) Calculate the mean rate of decrease in the lactic acid concentration in the blood between 10 minutes and 40 minutes. Give the unit. [2]

Question 19

supplement · 6 marks · 0610_T12_P1_20261013_Q19

Ethanol for use as a biofuel can be made by yeast. Sugar from sugar cane is dissolved in water and yeast is added. The mixture is kept in a large closed tank with no oxygen.

Topical Hub
topical-hub.pages.dev

(a) Write the balanced chemical equation for anaerobic respiration in yeast. [2]

(b) After several days, ethanol is produced more and more slowly, even though the temperature of the tank stays the same.

Suggest two reasons why. [2]

(c) Suggest why the tank is kept without oxygen. [2]

Question 20

supplement · 6 marks · 0610_T12_P1_20261013_Q20

Yeast can respire aerobically and anaerobically. A scientist sets up two flasks, X and Y. Each flask contains the same mass of yeast in the same volume of glucose solution at 25 °C.

- Air is bubbled through flask X.
- Flask Y is sealed and contains no oxygen.

Assume that the yeast in flask X respires only aerobically and the yeast in flask Y respires only anaerobically.

(a) State the number of molecules of carbon dioxide produced when one molecule of glucose is respired by the yeast in:

(i) flask X [1]

(ii) flask Y. [1]

(b) The yeast in each flask breaks down the same number of glucose molecules.

Calculate how many times more carbon dioxide molecules are produced in flask X than in flask Y. [1]

Topical Hub
topical-hub.pages.dev

(c) Name the product of respiration that is found in flask Y but is not produced by respiration in flask X. [1]

(d) The yeast in each flask releases the same amount of energy each hour.

Suggest why the glucose in flask Y is used up faster than the glucose in flask X. [2]

Topical Hub
topical-hub.pages.dev